US 20210171171A1

(19) **United States**

(12) **Patent Application Publication** (10) **Pub. No.: US 2021/0171171 A1**

HWANG et al. (43) **Pub. Date: Jun. 10, 2021**

(54) **SMALL UNDERWATER VEHICLE HAVING A HOVERING SYSTEM USING THE TUBE TYPE LAUNCHER AND METHOD FOR ASSEMBLING THE SAME**

(71) Applicant: **AGENCY FOR DEFENSE DEVELOPMENT**, Daejeon (KR)

(72) Inventors: **Kun Chul HWANG**, Daejeon (KR); **Dong Hoon LEE**, Daejeon (KR); **Jun Ho LEE**, Daejeon (KR); **Jae Moon YOON**, Daejeon (KR)

(73) Assignee: **AGENCY FOR DEFENSE DEVELOPMENT**, Daejeon (KR)

(21) Appl. No.: **17/106,582**

(22) Filed: **Nov. 30, 2020**

(30) **Foreign Application Priority Data**

Nov. 29, 2019 (KR) 10-2019-0156412

Publication Classification

(51) **Int. Cl.**

B63G 8/00	(2006.01)
B63G 8/08	(2006.01)
B63H 1/14	(2006.01)
B63H 21/12	(2006.01)

(52) **U.S. Cl.**

CPC ***B63G 8/001*** (2013.01); ***B63G 8/08*** (2013.01); *B63G 2008/004* (2013.01); ***B63H 21/12*** (2013.01); *B63G 2008/008* (2013.01); ***B63H 1/14*** (2013.01)

(57) **ABSTRACT**

An underwater vehicle having a hovering system using a tube type launcher. The underwater vehicle includes a streamlined body and a hovering system connected to a rear of the streamlined body to generate a kinetic force of the streamlined body. The hovering system includes an extension shaft extended to be connected to the rear, a connection assembly connected to the rear through the extension shaft, and an auxiliary propeller assembly connected to the connection assembly.

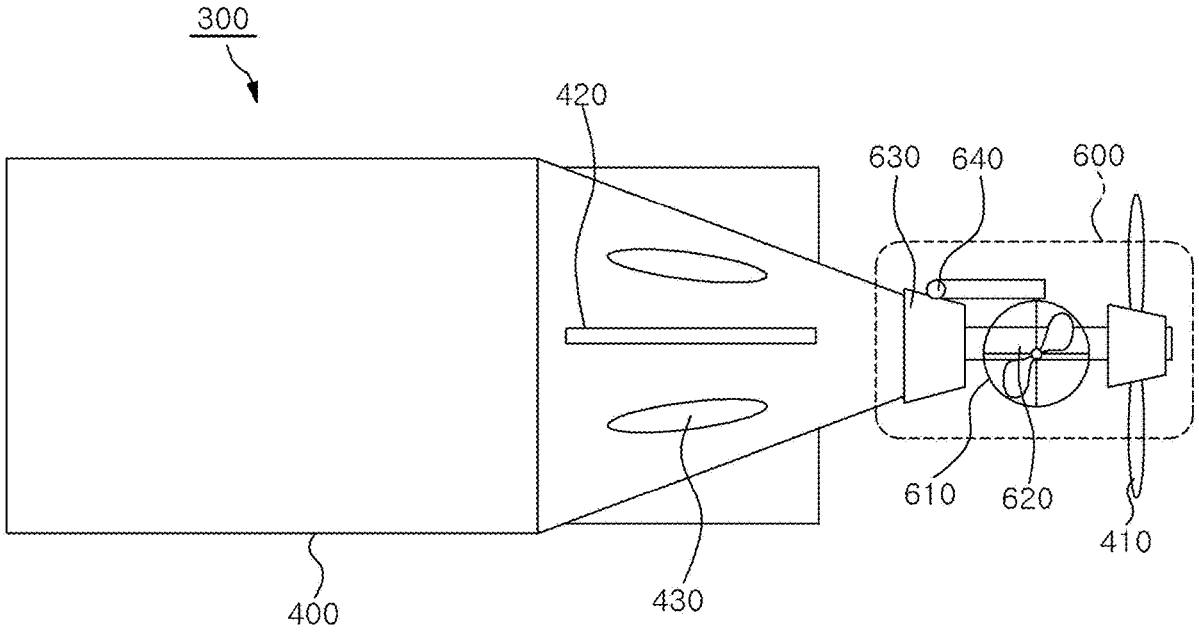


FIG. 1 *PRIOR ART*

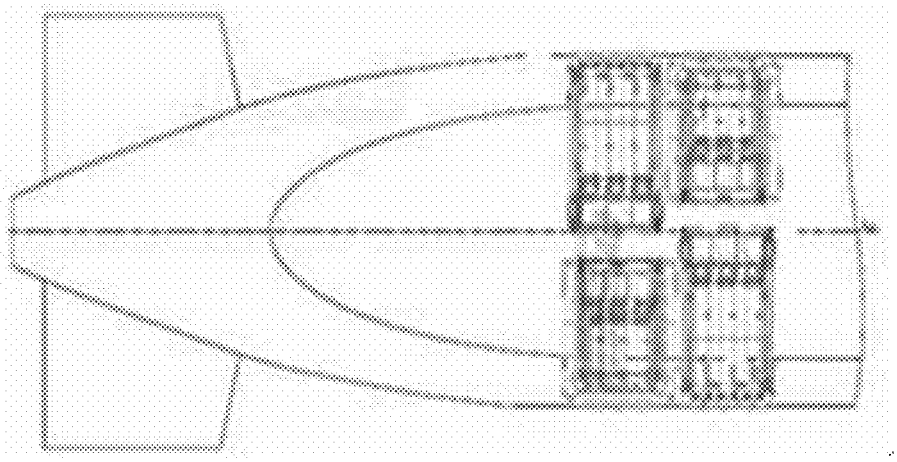


FIG. 2 *PRIOR ART*

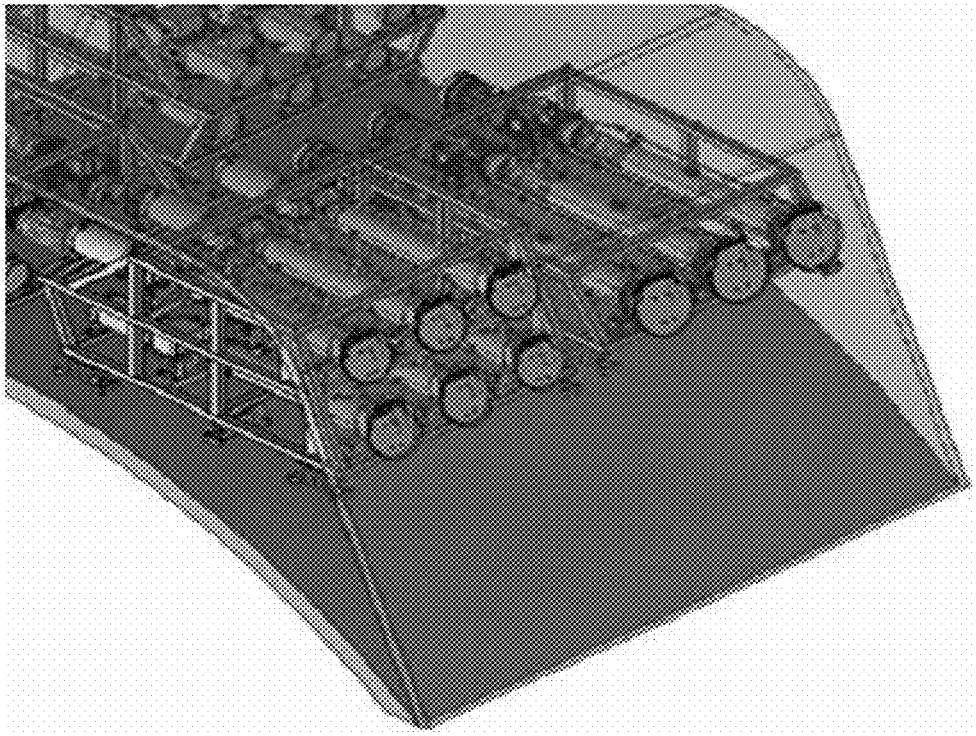


FIG. 3 *PRIOR ART*

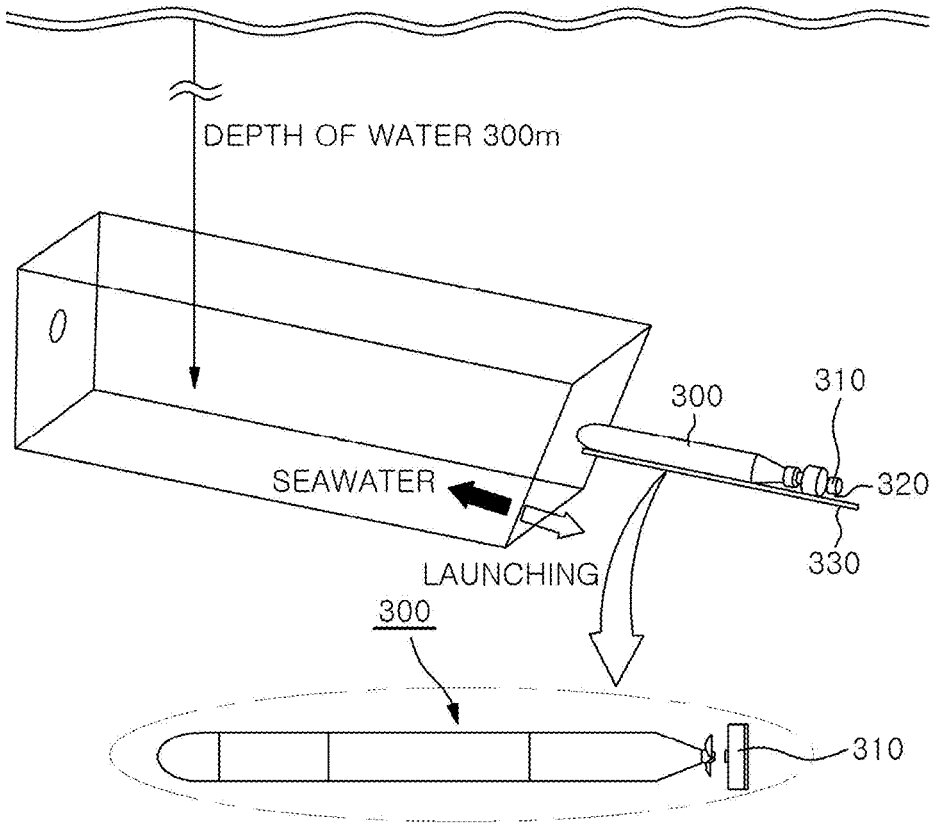


FIG. 4 *PRIOR ART*

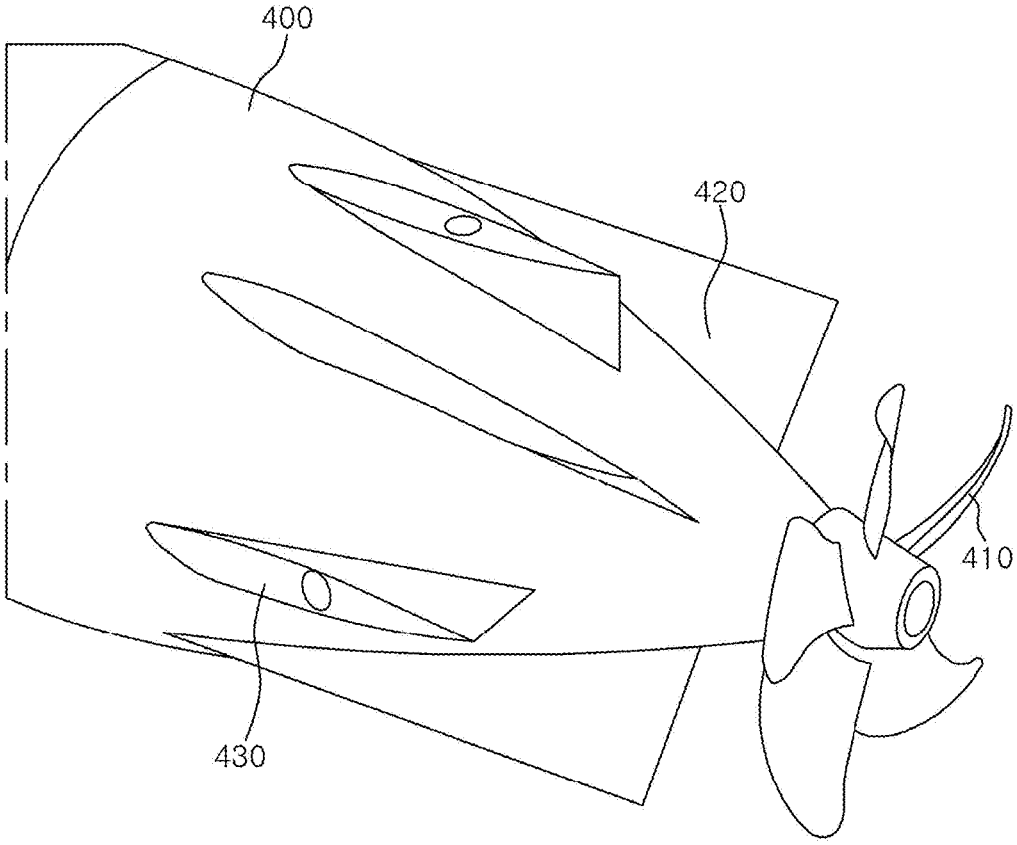


FIG. 5 *PRIOR ART*

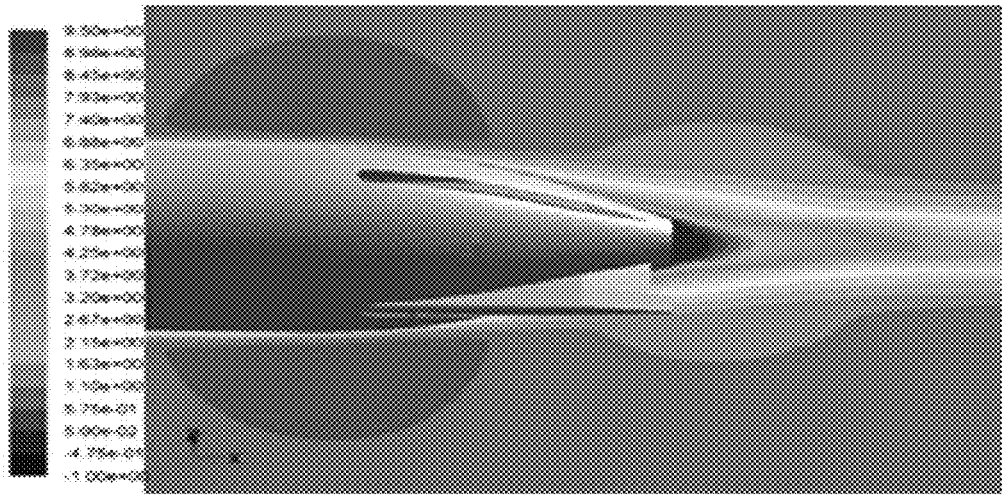


FIG. 6

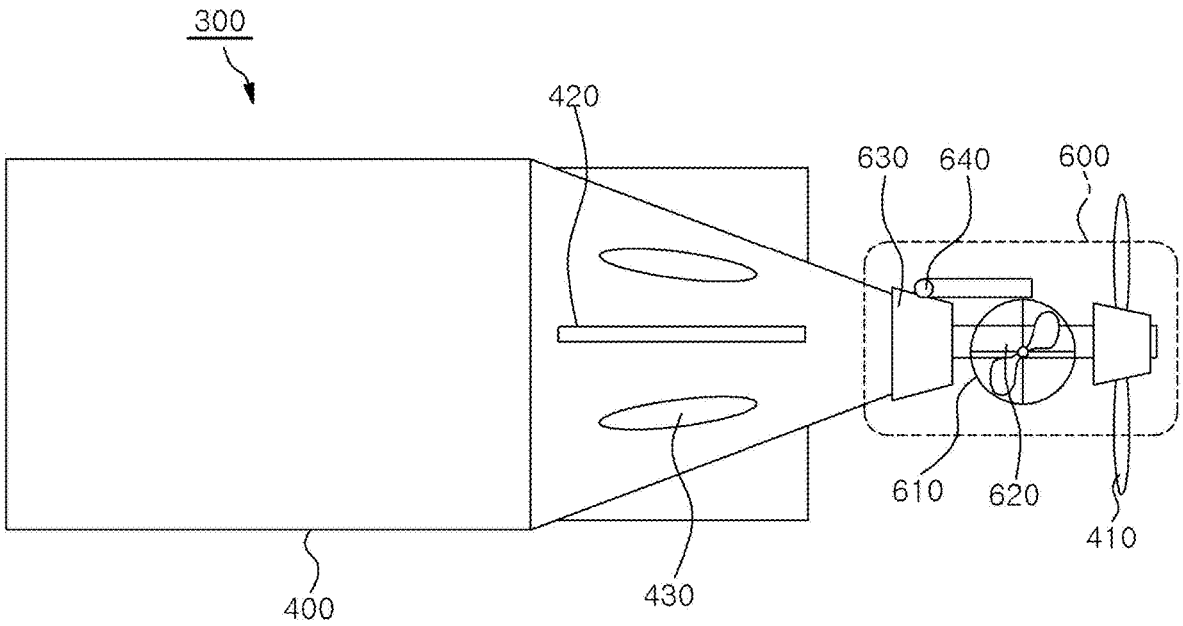


FIG. 7

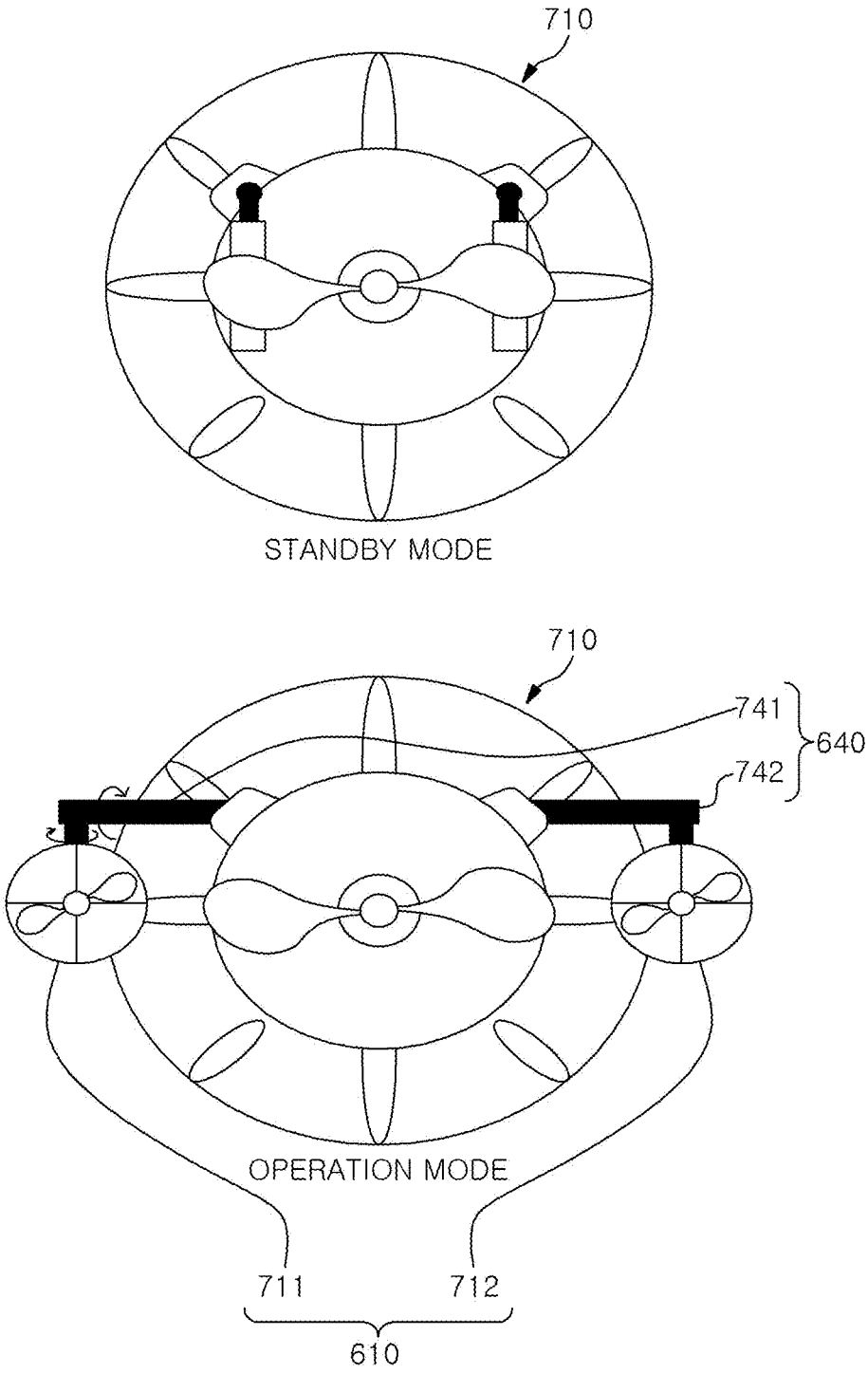


FIG.8

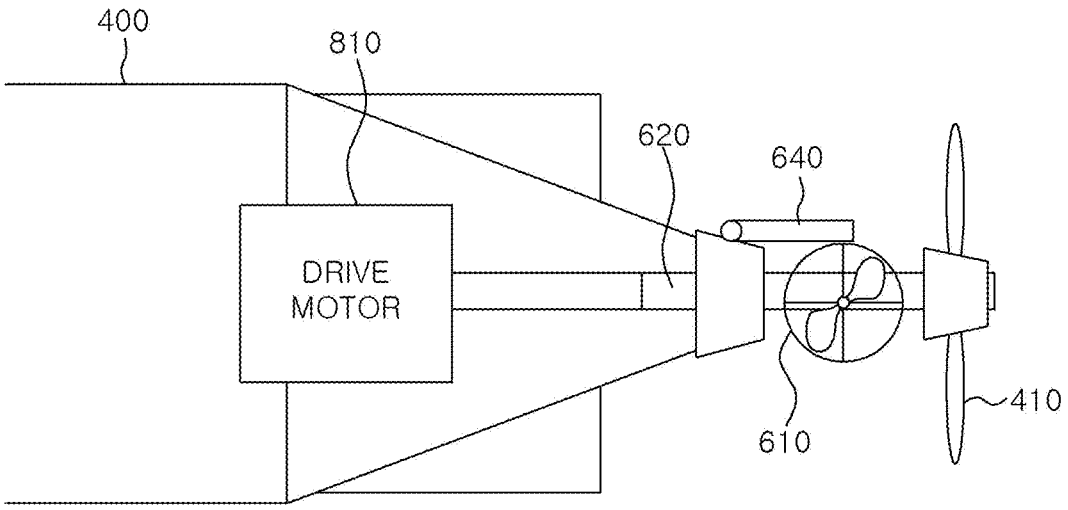


FIG. 9

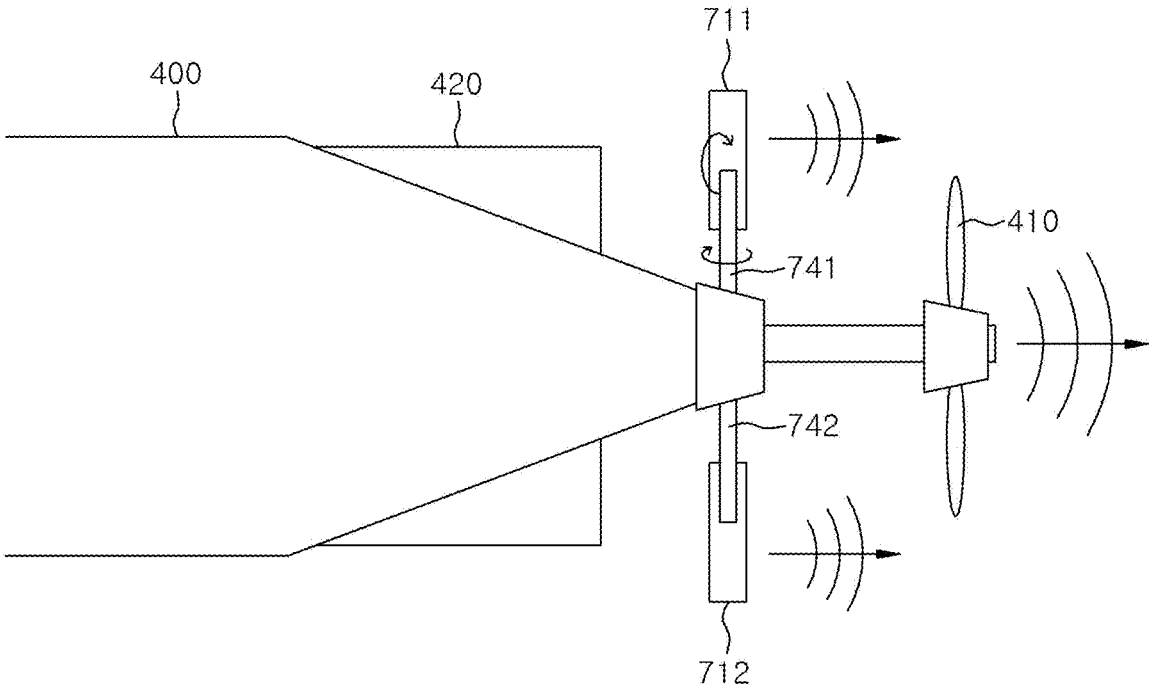


FIG. 10

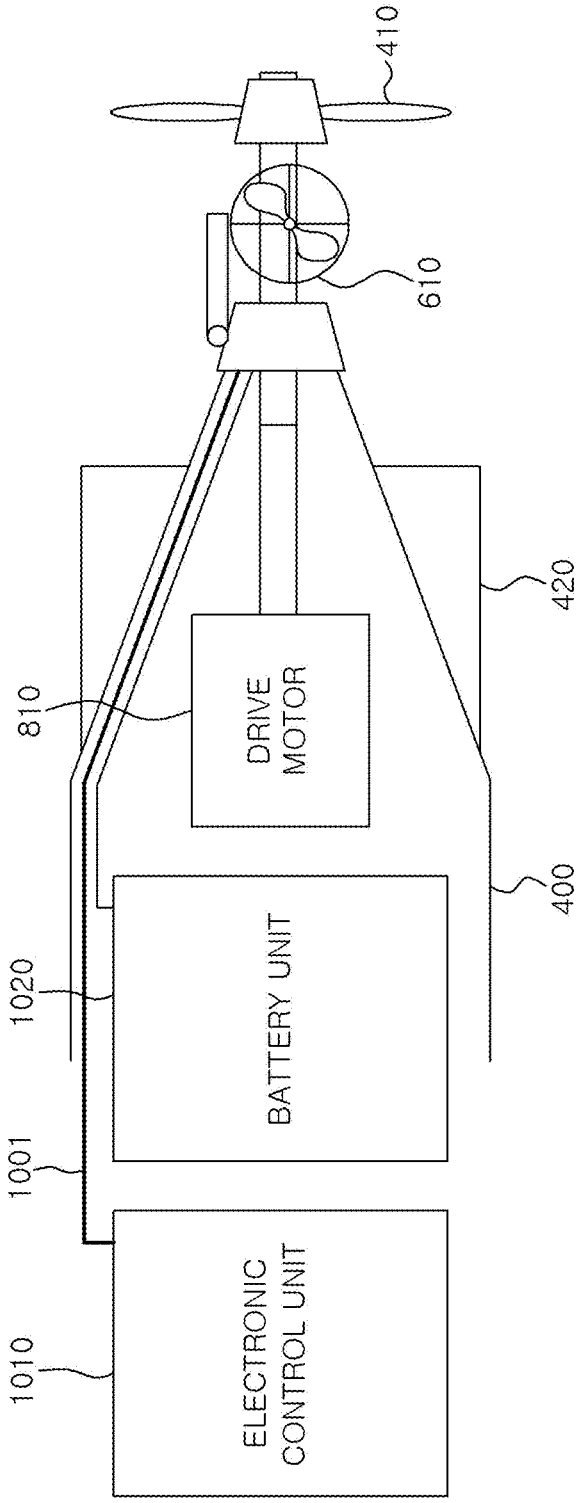

410
610
420
810
DRIVE MOTOR
400
1020
BATTERY UNIT
1001
1010
ELECTRONIC CONTROL UNIT

FIG. 11

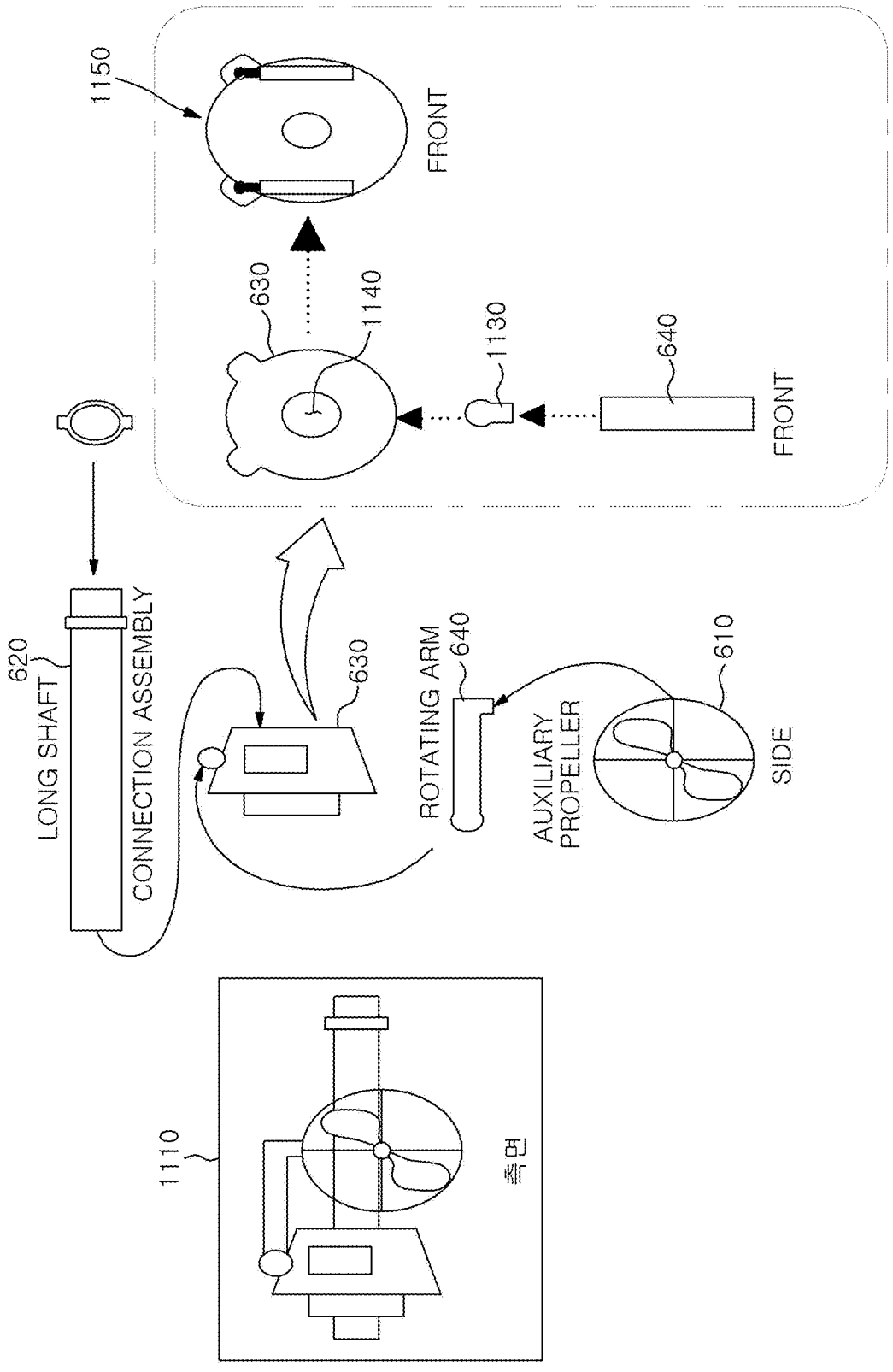

1150
FRONT
630
1140
1130
640
FRONT
620
LONG SHAFT
CONNECTION ASSEMBLY
630
ROTATING ARM
640
AUXILIARY PROPELLER
610
SIDE
1110
측면

FIG. 12

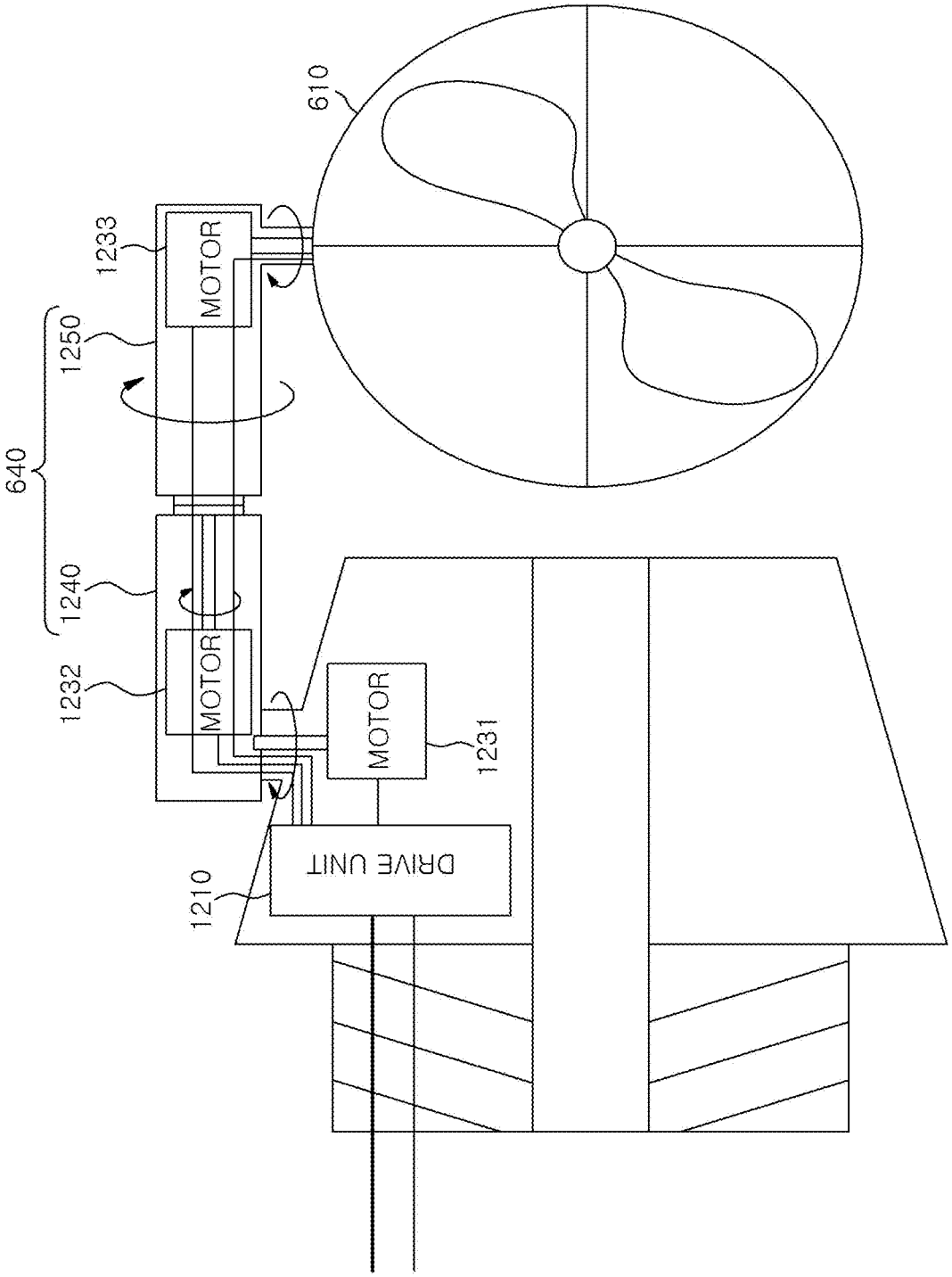

610
1233
MOTOR
1250
640
1240
1232
MOTOR
MOTOR
1231
DRIVE UNIT
1210

FIG. 13

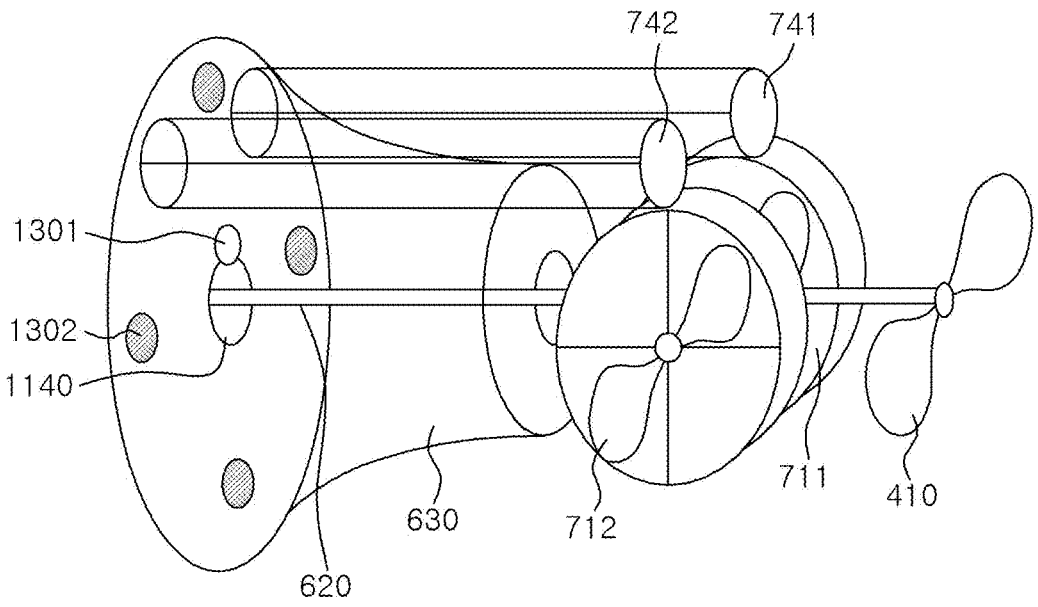


FIG. 14

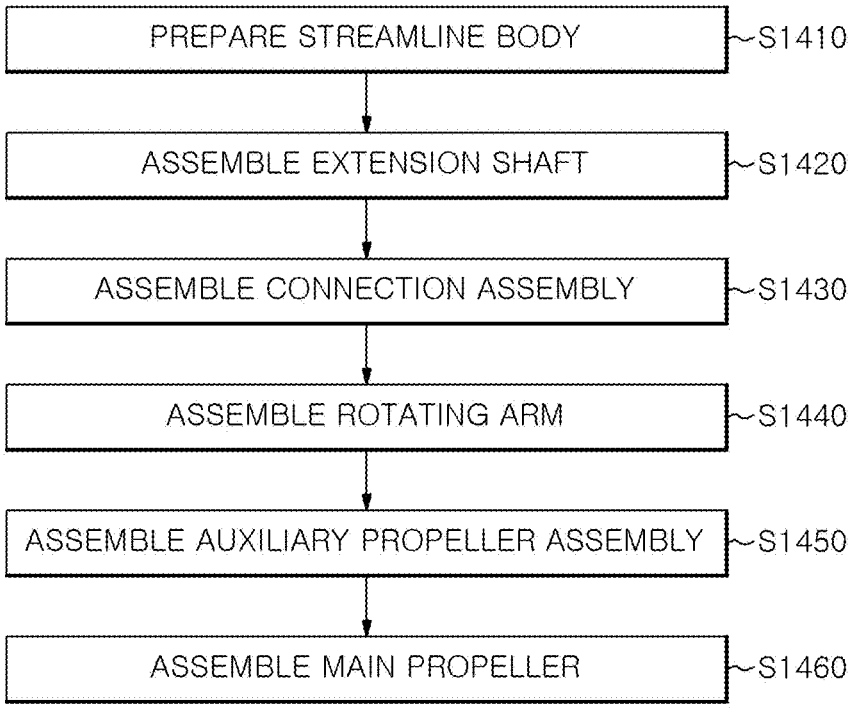
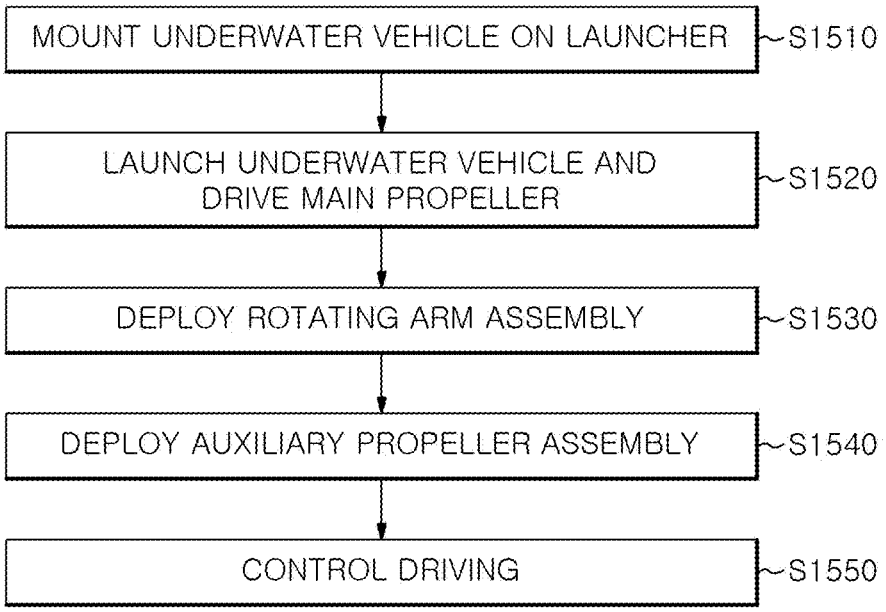

PREPARE STREAMLINE BODY
S1410
ASSEMBLE EXTENSION SHAFT
S1420
ASSEMBLE CONNECTION ASSEMBLY
S1430
ASSEMBLE ROTATING ARM
S1440
ASSEMBLE AUXILIARY PROPELLER ASSEMBLY
S1450
ASSEMBLE MAIN PROPELLER
S1460

FIG. 15


MOUNT UNDERWATER VEHICLE ON LAUNCHER
S1510
LAUNCH UNDERWATER VEHICLE AND DRIVE MAIN PROPELLER
S1520
DEPLOY ROTATING ARM ASSEMBLY
S1530
DEPLOY AUXILIARY PROPELLER ASSEMBLY
S1540
CONTROL DRIVING
S1550

SMALL UNDERWATER VEHICLE HAVING A HOVERING SYSTEM USING THE TUBE TYPE LAUNCHER AND METHOD FOR ASSEMBLING THE SAME

TECHNICAL FIELD

[0001] The present disclosure relates to a hovering technology of a small underwater vehicle, and more particularly, to a small underwater vehicle having a hovering system using a tube type launcher and a method for assembling the same.

BACKGROUND

[0002] FIG. **1** illustrates a layout view of a tube type launcher for launching a small underwater vehicle such as a torpedo decoy from an underwater platform such as a submarine. FIG. **2** is a perspective view of a starboard launcher illustrated in FIG. **1**. Underwater vehicles such as torpedoes, torpedo decoys, and unmanned submarines mostly take a dolphin-shaped streamlined structure, that is, a torpedo-shaped structure, to reduce fluid resistance in the water. A launching device (launcher) for such a torpedo-shaped underwater vehicle takes a tube type launcher shape of a cylindrical structure.

[0003] FIG. **3** is a layout view of the tube type launcher illustrated in FIG. **1**. A tube type launcher **330** uses a compressed air type launch method using a piston **320** or a ram piston type launch method using seawater, or takes a gas injection type launch method to improve the launch pressure. In the water **30**, a launcher of an underwater vehicle **300**, such as a torpedo or an unmanned submarine, usually employs a tube type launcher of the piston launch method. The underwater vehicle using the above-mentioned launcher is connected to a pressure head (pressure transmission unit) **310** that receives pressure transmitted from the piston of the launcher.

[0004] FIG. **4** illustrates a shape of a propeller unit of an underwater vehicle that includes a main propeller **410**, a control rudder **430**, and a stabilizing pin **420** to generate the kinetic force (propulsion force and control force) of a small underwater vehicle launched from a general tube type launcher. For an underwater vehicle **400** which is to be launched from the tube type launcher, in order to be mounted on the cylinder shaped launcher, a structure is not attached to the body thereof, and the control rudder and the propeller for generating the kinetic force of the underwater vehicle are attached to the tail part of the streamlined body **400**.

[0005] FIG. **5** is an experimental result graph showing a fluid flow (flow characteristics) around the underwater vehicle illustrated in FIG. **4**. For the underwater vehicle which is to be launched from the tube type launcher, in order to be mounted on the cylinder shaped launcher, a structure is not attached to the body thereof, and the control rudder and the propeller for generating the kinetic force of the underwater vehicle are attached to the tail part of the streamlined body.

[0006] In order to generate the kinetic force of an underwater vehicle of the torpedo shape, a propulsion force of a certain speed or higher has to be generated in the propeller. When a fluid flow (flow) is formed around the body of the underwater vehicle by the propulsion force of a certain speed or higher, drag (control force) that resist the flow is generated by operating the control rudder attached to the vehicle, which makes it possible to control the dynamic characteristics of the vehicle.

[0007] For the torpedo-shaped underwater vehicle, when the flow is not formed around the control rudder due to the low speed or the small size of the control rudder, it is difficult to control the direction of the vehicle, and thus, the underwater vehicle usually travels in the water at a certain speed or higher (vehicle speed) to obtain the control force.

[0008] In addition, for the torpedo-shaped underwater vehicle, most of the control rudders are arranged in the propeller unit at the rear of the vehicle, and accordingly, the height of the control rudder is usually adjusted depending on the radius of the vehicle. For a small underwater vehicle such as a torpedo decoy for submarines, the diameter of the vehicle is usually small, about 100 to 200 mm. Accordingly, the height of the control rudder has a small size of about 20 to 30 mm. As a result, for the small underwater vehicle ejected from the tube type launcher, such as the torpedo decoy for submarines, high-speed travelling of 15 kts (about 30 km/h) or more is usually conducted to obtain adequate control force.

[0009] In addition, since a small underwater vehicle using a tube type launcher needs to perform high-speed travelling in order to obtain the control force for the underwater movement, missions of low-speed travelling of 10 kts or less are very limited in execution, and propulsion energy source such as a battery is quickly lost due to high-speed travelling, which may result in a problem of short operating time of the vehicle. As a result, the small underwater vehicle using the tube type launcher has very limited mission performance due to problems of traveling speed and/or operating time.

PRIOR ART DOCUMENTS

[0010] (Patent Document 1) Korean Patent No. 10-0303379 (registered on Jul. 10, 2001)

[0011] (Patent Document 2) Korean Patent No. 10-1115211 (registered on Feb. 3, 2012)

[0012] (Patent Document 3) Japanese Patent No. 4417543 (registered on Dec. 4, 2009)

SUMMARY

[0013] The present disclosure has been suggested in order to solve the problem presented in the above Background, and an object thereof is to provide an underwater vehicle having a detachable hovering system to solve the problem of speed control of a small underwater vehicle launched from a tube type launcher, and a method for assembling the same.

[0014] Furthermore, another object of the present disclosure is to provide an underwater vehicle having a hovering system capable of compensating for an insufficient control force when the small underwater vehicle operates at a low speed by operating two auxiliary propellers horizontally and vertically when operating with an extendable auxiliary propeller system using rotating arms, and a method for assembling the same.

[0015] In order to achieve the above object, the present disclosure provides an underwater vehicle having a hovering system using a tube type launcher, the underwater vehicle including: a streamlined body; and a hovering system connected to a rear of the streamlined body to generate a kinetic force of the streamlined body.

[0016] Further, the hovering system may include: an extension shaft extended to be connected to the rear; a connection assembly connected to the rear through the extension shaft; and an auxiliary propeller assembly connected to the connection assembly.

[0017] Further, the hovering system may include a main propeller fastened to an end of the extension shaft.

[0018] Further, the underwater vehicle may further include a rotating arm assembly, wherein the auxiliary propeller assembly is connected to the connection assembly through the rotating arm assembly to be rotatable at a predetermined angle.

[0019] Further, the auxiliary propeller assembly may include a first auxiliary propeller and a second auxiliary propeller.

[0020] Further, the first auxiliary propeller and the second auxiliary propeller may move in one of up, down, right, and left directions.

[0021] Further, each of the first auxiliary propeller and the second auxiliary propeller may be rotatable within a range of 0° to 360°.

[0022] Further, the rotating arm assembly may include a first rotating arm and a second rotating arm such that each of the first auxiliary propeller and the second auxiliary propeller is connected to the connection assembly.

[0023] Further, the first rotating arm and the second rotating arm may be rotated left and right, respectively, such that an angle between the first rotating arm and the second rotating arm is changeable from 0° up to 180°.

[0024] Further, each of the first rotating arm and the second rotating arm may include: a first rotation member connected to a first sub-motor installed inside the connection assembly; a second rotation member connected to the first rotation member; a second sub-motor installed inside the first rotation member to rotate the second rotation member; and a third sub-motor installed inside the second rotation member to rotate the first auxiliary propeller and the second auxiliary propeller.

[0025] Further, the second rotation member may rotate within a range of 0° to 360°.

[0026] Further, the connection assembly may be detachably connected to the rear by bolting.

[0027] On the other hand, another embodiment of the present disclosure provides a method for assembling an underwater vehicle having a hovering system using a tube type launcher, including: preparing a streamlined body; and connecting the hovering system generating a kinetic force of the streamlined body to a rear of the streamlined body.

[0028] According to the present disclosure, by improving the travelling ability of the small underwater vehicle using the tube type launcher, it is possible to not only enable a low-speed travelling but also increase a speed of a high-speed travelling by generating an auxiliary propulsion force in some cases.

[0029] In addition, as another effect of the present disclosure, the improvement in travelling ability improves the mission performance capability of the small underwater vehicle using the existing tube type launcher, which is limited due to the high-speed travelling mode.

[0030] In addition, as another effect of the present invention, it is possible to expect a variety of missions such as landing, search, and attack of the small underwater vehicle using the tube type launcher.

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] FIG. **1** is a layout view of a general tube type launcher.

[0032] FIG. **2** is a perspective view of a starboard launcher illustrated in FIG. **1**.

[0033] FIG. **3** is a layout view of the tube type launcher illustrated in FIG. **1**.

[0034] FIG. **4** illustrates a shape of a propeller unit of the underwater vehicle launched from the tube type launcher illustrated in FIG. **1**.

[0035] FIG. **5** is an experimental result graph showing a fluid flow (flow characteristics) around the underwater vehicle illustrated in FIG. **4**.

[0036] FIG. **6** is a schematic view of an underwater vehicle having a hovering system using a tube type launcher according to an embodiment of the present disclosure.

[0037] FIG. **7** is a rear view of the hovering system illustrated in FIG. **6** viewed from the rear to the front.

[0038] FIG. **8** is a conceptual view for driving the hovering system illustrated in FIG. **6**.

[0039] FIG. **9** is a conceptual view showing an operating method for the hovering system illustrated in FIG. **6**.

[0040] FIG. **10** is a diagram of connecting lines of the hovering system illustrated in FIG. **6**.

[0041] FIG. **11** is a conceptual view for assembling the hovering system illustrated in FIG. **6**.

[0042] FIG. **12** is a diagram showing the configuration of a connection assembly and a rotating arm of the hovering system illustrated in FIG. **6**.

[0043] FIG. **13** is a diagram showing a perspective view of the hovering system illustrated in FIG. **6**.

[0044] FIG. **14** is a flowchart showing a process of assembling a hovering system according to the embodiment of the present disclosure to a streamlined body.

[0045] FIG. **15** is a flowchart showing a process of controlling operations of the hovering system assembled according to FIG. **14**.

DETAILED DESCRIPTION

[0046] The present disclosure may have various modifications, and may be implemented as various embodiments, and thus specific embodiments will be illustrated in the Drawings and described in detail in the Detailed Description. However, it is not intended to limit the present disclosure to specific embodiments, and it is to be understood to include all changes, equivalents, and substitutes included in the spirit and scope of the present disclosure.

[0047] Similar reference numerals are used to indicate similar elements throughout drawings. Terms including "first", "second", and the like may be used to describe various elements, but the elements are not limited by the terms. The terms are used to only distinguish one element from the other element.

[0048] For example, without departing from the scope of the present disclosure, a first element may be referred to as a second element, and similarly, a second element may be referred to as a first element. The term "and/or" includes a combination of a plurality of related stated items or any of a plurality of related stated items.

[0049] Unless otherwise defined, all terms used herein, including technical and scientific terms, have the same meaning as commonly understood by those of ordinary skill in the art to which the present disclosure belongs.

[0050] Terms such as those defined in a commonly used dictionary are to be interpreted as having a meaning consistent with the meaning in the context of the related technology, and are not to be interpreted in an ideal or excessively formal meaning unless explicitly defined in the present application.

[0051] Hereinafter, an underwater vehicle having a hovering system using a tube type launcher according to an embodiment of the present disclosure and a method for assembling the same will be described in detail with reference to the accompanying drawings.

[0052] In general, underwater vehicles such as torpedoes, torpedo decoys, and unmanned submarines mostly take a dolphin-shaped streamlined structure, that is, a torpedo-shaped structure, to reduce fluid resistance in the water. A launching device (launcher) for such a torpedo-shaped underwater vehicle takes a tube type launcher shape of a cylindrical structure.

[0053] A tube type launcher uses a compressed air type launch method using a piston or a ram piston type launch method using seawater, or takes a gas injection type launch method to improve the launch pressure. A launcher of an underwater vehicle, such as a torpedo or an unmanned submarine, usually employs a tube type launcher of a piston launch method.

[0054] For the underwater vehicle which is to be launched from the tube type launcher, in order to be mounted on the launcher of the cylindrical cylinder shape, a structure is not attached to the body thereof, and the control rudder and the propeller for generating the kinetic force of the underwater vehicle are attached to the tail part of the streamlined body.

[0055] FIG. **5** is an experimental result graph showing a fluid flow (flow characteristics) around the underwater vehicle illustrated in FIG. **4**. The flow characteristics formed by the launching of the underwater vehicle show that the flow is small in the region immediately behind a nozzle of the underwater vehicle, and is convex upward and downward in a position spaced a certain distance from a rear end portion where the main propeller is positioned, which is a result of expansion of the flow region by the main propeller. The position of the auxiliary propellers described in the present disclosure is located between the rear end portion and the main propeller, and for this purpose, the position of the main propeller is located further behind in a case where there is no auxiliary propeller, and the auxiliary propellers uses an extension shaft **620** in which the length thereof is extended.

[0056] FIG. **6** is a schematic view of an underwater vehicle **300** having a hovering system **600** using a tube type launcher according to the embodiment of the present disclosure. Referring to FIG. **6**, the underwater vehicle **300** may include a streamlined body **400**, the hovering system **600** connected to the rear of the streamlined body **400**, and the like.

[0057] At the rear of the streamlined body **400**, the control rudder **430**, and the stabilizing pin **420** are formed. Four stabilizing pins **420** are arranged in a cross shape, and the control rudder **430** is formed between the stabilizing pins **420**. The height of the control rudder **430** is approximately 20 to 30 mm. When a fluid flow (flow) is formed around the body of the underwater vehicle **300** by the propulsion force of a certain speed or higher, drag (control force) that resist the flow is generated by operating the control rudder **430** attached to the streamlined body **400** of the underwater vehicle **300**, which makes it possible to control the dynamic characteristics of the underwater vehicle **300**.

[0058] For the underwater vehicle **400** launched from the tube type launcher, the tube type launcher generally has a cylindrical cylinder shape. Therefore, in order to be mounted on the cylindrical cylinder shape, a structure is not attached to the streamlined body **400**, and the control rudder **430** and the propellers **410** and **610** for generating the kinetic force of the underwater vehicle are attached to the tail part of the streamlined body **400**.

[0059] The hovering system **600** may be configured to include the extension shaft **620** extended to be connected to the rear of the streamlined body **400**, the main propeller **410** assembled by being fastened to an end of the extension shaft **620**, the connection assembly **630** assembled by being connected to the rear of the streamlined body **400** through the extension shaft **620**, and the auxiliary propeller assembly **610** assembled by being connected to the connection assembly **630** through the rotating arm assembly **640**.

[0060] The kinetic force may include the propulsion force and the control force. Among them, the propulsion force is generated by the main propeller **410**, and the control force for the direction is generated by the auxiliary propeller assembly **610**.

[0061] FIG. **7** is a rear view of the hovering system **600** illustrated in FIG. **6** viewed from the rear to the front. Referring to FIG. **7**, the auxiliary propeller assembly **610** includes a first auxiliary propeller **711** and a second auxiliary propeller **712**, and the rotating arm assembly **640** includes a first rotating arm **741** and a second rotating arm **742**. The first rotating arm **741** and the second rotating arm **742** function to connect the first auxiliary propeller **711** and the second auxiliary propeller **712** to the connection assembly **630**, respectively, for assembly.

[0062] By simultaneously operating the two auxiliary propellers **711** and **712** up, down, right, and left through the rotating arms **741** and **742**, the insufficient control force when the underwater vehicle **300** operates at a low speed is compensated. That is, the two auxiliary propellers **711** and **712** move in any one of the up, down, right and left directions, thereby compensating for the control force for the direction.

[0063] With continued reference to FIG. **7**, the two auxiliary propellers **711** and **712** are changed from a standby mode **710**, which is a state before being extended, to an operation mode **720**, which is a state after being extended. In the operation mode, the two rotating arms **741** and **742** may be rotated left and right, respectively such that an angle therebetween changes up to 180°. In addition, the two rotating arms **741** and **742** may be rotated left and right up to 90°, respectively.

[0064] Unlike this, the first auxiliary propeller **711** and/or the second auxiliary propeller **712** may move in any one of up, down, right, and left directions. In addition, the first auxiliary propeller **711** and/or the second auxiliary propeller **712** may rotate 360° up, down, right, and left.

[0065] Therefore, it is possible not to change the design shape of the existing small underwater vehicle using the tube type launcher. The hovering system is attached to the rear of the underwater vehicle **300** by replacing the existing propeller connection portion. The connection assembly **630** (shown in FIG. **6**) attached to the rear is connected to the rotating arms **741** and **742** driving the two auxiliary propellers **711** and **712**. In addition, the connection assembly **630**

has a shaft hole in the center to pass the extension shaft **620** connecting the existing propeller and a propulsion motor of the underwater vehicle.

[0066] When the extended length of the auxiliary propellers, including the length of the first rotating arm **741** and the second rotating arm **742** supporting the auxiliary propellers **711** and **712** and the width of the connection assembly **630** therebetween, is equal to or greater than the diameter of the main propeller **410**, flow interference between the auxiliary propellers **711** and **712** and the main propeller **410** may be avoided. In addition, the lengths of the first rotating arm **741** and the second rotating arm **742** of the auxiliary propeller have to be smaller than the length of the extension shaft **620**. In addition, when the auxiliary propellers **711** and **712** are extended, the central axis of the auxiliary propellers **711** and **712** and the central axis of the main propeller **410** may be parallel.

[0067] FIG. **8** is a conceptual view for driving the hovering system **600** illustrated in FIG. **6**. Referring to FIG. **8**, one end of the extension shaft **620** is assembled by being connected to a drive shaft of the drive motor **810**, and the other end of the extension shaft **620** is assembled by being connected to the main propeller **410**. The drive motor **810** is rotated to rotate the extension shaft **620**.

[0068] FIG. **9** is a conceptual view showing an operating method for the hovering system **600** illustrated in FIG. **6**. Referring to FIG. **9**, the hovering system **600** is an auxiliary mounting system for enabling low-speed traveling of the underwater vehicle **300** without changing the shape of the propeller unit at the rear of the existing underwater vehicle **300**.

[0069] Therefore, the hovering system **600** is mounted on the existing tube type launcher as it is, and after being launched from the tube type launcher, it functions to generate the auxiliary control force required for low-speed traveling through the two propellers **711** and **712** by receiving a low-speed traveling command from the small underwater vehicle **300**. In the hovering system **600**, the rotating arms **741** and **742** stand by in a folded form inside the tube type launcher, and after being launched, the rotating arms **741** and **742** are extended according to the traveling command. Accordingly, by the auxiliary propellers **711** and **712** operating, auxiliary propulsion force and/or control force are transmitted to the underwater vehicle **300**.

[0070] FIG. **9** shows that the auxiliary propeller assemblies **711** and **712** parallel to the extension shaft **620** are extended to be perpendicular to the extension shaft **620** and then the auxiliary propellers **711** and **712** are rotated.

[0071] In conclusion, the hovering system **600** according to the embodiment of the present disclosure improves the travelling ability of the underwater vehicle **300** by using the tube type launcher, thereby making it possible to enable low-speed travelling, and furthermore, in some cases, to increase a high-speed travelling speed by generating the auxiliary propulsion force. This improvement in travelling ability may improve the mission performance capability of the small underwater vehicle using the existing tube type launcher, which is limited due to the high-speed travelling mode. Therefore, this makes it possible to perform various missions such as landing, search, and attack of the small underwater vehicle using the tube type launcher.

[0072] FIG. **10** is a diagram of connecting lines of the hovering system **600** illustrated in FIG. **6**. Referring to FIG. **10**, the hovering system **600** is connected, by a line, to an electronic control unit **1010** and a battery unit **1020** configured inside the underwater vehicle **300**. The hovering system **600** receives motion information, low speed or high speed travelling command information of the underwater vehicle **300** through a signal connection line **1001** and the electronic control unit **1010**, and then, generates, from the auxiliary propellers **711** and **712**, enough kinetic force to satisfy it. The hovering system **600** receives power required for driving the auxiliary propellers **711** and **712** through a power connection line with the battery unit **1020**.

[0073] The electronic control unit **1010** may include a microprocessor, a circuit device, a memory, a program, and the like. The program has an algorithm for generating driving information in advance using information acquired through a sensor, information input in advance, and the like. The electronic control unit **1010** may function to receive the motion information and the travelling command from the vehicle, generate driving information of the auxiliary propeller, and transmit the generated driving information.

[0074] As the battery unit **1020**, a CR2032 lithium battery, a general battery, or the like, may be used. Of course, a rechargeable secondary battery may also be used.

[0075] FIG. **11** is a conceptual view for assembling the hovering system illustrated in FIG. **6**. Referring to FIG. **11**, the connection assembly **630** has a shaft hole **1140** in the center through which the extension shaft **620** passes. A fixing ring **1120** is inserted and fixed to one end of the extension shaft **620**. The fixing ring **1120** protrudes on the surface of the extension shaft **620** so that the main propeller **410** is not inserted to a certain degree or more. The connection assembly **630** is attached to the rear end portion of the underwater vehicle **300** through a connection bolt hole (not illustrated).

[0076] In addition, at the top of the connection assembly **630**, the rotating arm assembly **640** is assembled by being connected through a frame **1130** of a gear. A front view **1150** thereof is illustrated in FIG. **11**. An auxiliary propeller assembly **610** is assembled by being connected to the rotating arm assembly **640**.

[0077] FIG. **12** is a diagram showing the configuration of the connection assembly **630** and the rotating arm assembly **640** of the hovering system **600** illustrated in FIG. **6**. Referring to FIG. **12**, the rotating arm assembly **640** includes a first rotation member **1240** and a second rotation member **1250**. The first rotation member **1240** is assembled by a frame of a gear and a shaft of a first sub-motor **1231**.

[0078] Put more precisely, the frame **1130** of the gear is installed on the first rotation member **1240** so as to mesh with the shaft (not illustrated) of the first sub-motor **1231**. That is, the shaft end of the first sub-motor **1231** may assembled by being inserted into the frame **1130** of the gear. In this case, several through holes (not illustrated) are formed on the surface of the frame **1130** of the gear for fixing, and screw bolts (not illustrated) may be fastened through the through holes. Accordingly, when the first sub-motor **1231** rotates between 0° and 90°, the first rotation member **1240** also rotates between 0° and 90°.

[0079] The first rotation member **1240** and the second rotation member **1250** may be coupled in a similar manner as above. Of course, the second sub-motor **1232** is installed inside the first rotation member **1240** to rotate the second rotation member **1250**. Accordingly, when the second sub-motor **1232** rotates between 0° and 360°, the second rotation member **1250** also rotates between 0° and 360°.

[0080] The second rotation member **1250** and the auxiliary propeller assembly **610** may be coupled in a similar manner as above. Of course, a third sub-motor **1233** is installed inside the second rotation member **1250** to rotate the auxiliary propeller assembly **610**. Accordingly, when the third sub-motor **1233** rotates between 0° and 360°, the auxiliary propeller assembly **610** also rotates right and left between 0° and 360°.

[0081] Through the extension shaft **620**, the main propeller **410** and the driving motor **810** (shown in FIG. **8**) inside the underwater vehicle **300** are connected. In addition, a driver **1210** includes a circuit for driving the first to third sub-motors **1231** to **1233**.

[0082] FIG. **13** is a diagram showing a perspective view of the hovering system **600** illustrated in FIG. **6**. Referring to FIG. **13**, connection bolt holes **1302** are formed in the connection assembly **630**, and through the connection bolt holes **1302**, the connection assembly **630** is assembled by being connected to the rear of the streamlined body **400** (shown in FIG. **4**) in a bolting manner so that it is detachable.

[0083] In addition, signal wiring and/or power wiring inside the underwater vehicle **300** are made through a connection line hole **1301**.

[0084] In addition, the first rotating arm **741** and the second rotating arm **742** are simply shown, for understanding, in FIG. **13**, but it includes the first rotation member **1240**, the second rotation member **1250**, the frame of the gear, the sub-motors **1231** to **1233**, the shafts of the sub-motors, and the like.

[0085] FIG. **14** is a flowchart showing a process of assembling a hovering system according to the embodiment of the present disclosure to the streamlined body **400**. Referring to FIG. **14**, the streamlined body **400** is prepared, and the extension shaft **620** is assembled at the rear of the streamlined body **400** (steps S**1410** and S**1420**). Then, the connection assembly **630** is assembled by being fastened to the rear of the streamlined body **400** (step S**1430**). Then, one end of the rotating arm assembly **640** is assembled by being fastened to the upper front of the connection assembly **630**, and the auxiliary propeller assembly **610** is assembled by being fastened to the other end of the rotating arm assembly **640** (step S**1440**). Then, the main propeller **410** is assembled by being fastened to the end of the extension shaft **620**. Of course, the assembly order is an example for understanding, and some orders may be changed.

[0086] FIG. **15** is a flowchart showing a process of controlling operations of the hovering system assembled according to FIG. **14**. Referring to FIG. **15**, the assembled underwater vehicle **300** is mounted on a tube type launcher (step S**1510**). Then, when the underwater vehicle **300** is launched, the main propeller **410** is driven accordingly (step S**1520**).

[0087] Then, the rotating arm assembly **640** is extended for direction change or the like according to a preset programmed command or an external adjustment command (step S**1530**). The extension may be performed such that the auxiliary propeller assemblies **711** and **712** parallel to the extension shaft **620** are perpendicular to the extension shaft **620**. Of course, the angle of extension may be different depending on the programmed command or external adjustment command.

[0088] Then, the auxiliary propeller assemblies **711** and **712** are extended (step S**1540**). As described above, the extension is performed by rotating up and down between 0° and 360°, and then rotating left and right between 0° and 360°. Of course, it is also possible to rotate left and right first and then rotate up and down.

[0089] Then, the auxiliary propeller assemblies **711** and **712** are controlled to be driven (step S**1550**).

What is claimed is:

1. An underwater vehicle having a hovering system using a tube type launcher, the underwater vehicle comprising:
- a streamlined body; and
- a hovering system connected to a rear of the streamlined body to generate a kinetic force of the streamlined body.

2. The underwater vehicle of claim **1**, wherein the hovering system includes:
- an extension shaft extended to be connected to the rear;
- a connection assembly connected to the rear through the extension shaft; and
- an auxiliary propeller assembly connected to the connection assembly.

3. The underwater vehicle of claim **2**, wherein the hovering system includes a main propeller fastened to an end of the extension shaft.

4. The underwater vehicle of claim **2**, further comprising a rotating arm assembly,
- wherein the auxiliary propeller assembly is connected to the connection assembly through the rotating arm assembly to be rotatable at a predetermined angle.

5. The underwater vehicle of claim **4**, wherein the auxiliary propeller assembly includes a first auxiliary propeller and a second auxiliary propeller.

6. The underwater vehicle of claim **5**, wherein the first auxiliary propeller and the second auxiliary propeller move in one of up, down, right, and left directions.

7. The underwater vehicle of claim **5**, wherein each of the first auxiliary propeller and the second auxiliary propeller is rotatable within a range of 0° to 360°.

8. The underwater vehicle of claim **5**, wherein the rotating arm assembly includes a first rotating arm and a second rotating arm such that each of the first auxiliary propeller and the second auxiliary propeller is connected to the connection assembly.

9. The underwater vehicle of claim **8**, wherein the first rotating arm and the second rotating arm are rotated left and right, respectively, such that an angle between the first rotating arm and the second rotating arm is changeable from 0° up to 180°.

10. The underwater vehicle of claim **8**, wherein each of the first rotating arm and the second rotating arm includes:
- a first rotation member connected to a first sub-motor installed inside the connection assembly;
- a second rotation member connected to the first rotation member;
- a second sub-motor installed inside the first rotation member to rotate the second rotation member; and
- a third sub-motor installed inside the second rotation member to rotate the first auxiliary propeller and the second auxiliary propeller.

11. The underwater vehicle of claim **10**, wherein the second rotation member rotates within a range of 0° to 360°.

12. The underwater vehicle of claim **2**, wherein the connection assembly is detachably connected to the rear by bolting.

13. A method for assembling an underwater vehicle having a hovering system using a tube type launcher, the method comprising:

preparing a streamlined body; and

connecting the hovering system generating a kinetic force of the streamlined body to a rear of the streamlined body.

* * * * *